🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.3

Ratio Reasoning: Convert Measurements Between Systems

Lesson 3-7 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can solve real-world rate problems by finding and using unit rates.

Language Objective: I can explain my solution using the words rate, unit rate, per, and proportion.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Ratio and Rate Problem Solving | Rate | Unit rate | Per | Problem solving | Proportion | Ratio table

👆 *Tap any word to see what it means and a picture.*

1. ________ — Use ratios and rates to compare quantities and solve real-world problems.

2. A ratio comparing two quantities with different units is a ________ .

3. A rate for exactly one unit, like $3 per 1 pound, is a ________ .

4. The word that means 'for each' in a rate, like miles ________ hour, is ________ .

5. Using a plan and math steps to find an answer to a real situation is ____________________.

6. An equation that two ratios are equal is a ____________________.

7. A table that lists equivalent ratios in rows or columns is a ____________________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

COMPARE

A runner trains for a 5K race. During practice, she runs 3 miles in 27 minutes. She wants to know her pace (minutes per mile) and predict how long the full 5K (3.1 miles) will take at the same rate?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Ratio and Rate Problem Solving** — Use ratios and rates to compare quantities and solve real-world problems.
  Resolución de problemas de razones y tasas — Usar razones y tasas para comparar cantidades y resolver problemas del mundo real.

- [ ] **Rate** — A ratio comparing two amounts with different units, like miles per hour.
  Tasa — Una razón que compara dos cantidades con unidades distintas, como millas por hora.

- [ ] **Unit rate** — A rate for just 1 of something, like cost for 1 item.
  Tasa unitaria — Una tasa para solo 1 de algo, como el precio de 1 artículo.

- [ ] **Per** — For each one. Example: 5 dollars per book.
  Por — Por cada uno. Ejemplo: 5 dólares por libro.

- [ ] **Problem solving** — Using ratios and rates to find a missing amount.
  Resolución de problemas — Usar razones y tasas para encontrar una cantidad que falta.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

She runs 3 miles in 27 minutes, so her pace is ____ minutes per mile and the 3.1-mile 5K takes about ____ minutes.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Divide the total cost by the number of pounds to find the cost per 1 pound.

Evidence: Students divide $18 by 3 to get $6 per pound and explain the unit rate lets them find the cost of any number of pounds.

Reasoning: This evidence connects the definition of ratio and rate problem solving to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **She runs 3 miles in 27 minutes, so her pace is ____ minutes per mile and the 3.1-mile 5K takes about ____ minutes.**
- **My answer is ____.**

  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **I know because ____.**

  Lo sé porque ____.
- **So ____.**

  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Ratio and Rate Problem Solving
2. **Notes 2:** Rate
3. **Notes 3:** Unit rate
4. **Notes 4:** Per
5. **Notes 5:** Problem solving
6. **Notes 6:** Proportion
7. **Notes 7:** Ratio table
8. **Watch for this mistake:** *A common mistake in Ratio and Rate Problem Solving is dividing in the wrong order when setting up the unit rate — for example, finding "minutes per smoothie" (6 ÷ 18) instead of "smoothies per minute" (18 ÷ 6) — or using addition instead of multiplication to scale the unit rate, like adding $0.75 + 10 instead of multiplying $0.75 × 10. Before you submit, check: does your unit rate match the units the question is asking for, and did you multiply (not add) to scale it to the new amount?*
9. **Try It 1:** A. 28 carrots — *Unit rate: 8 carrots ÷ 2 minutes = 4 carrots per minute. In 7 minutes: 4 × 7 = 28 carrots.*
10. **Try It 2:** A. 42 ounces — *Unit rate: 18 ounces ÷ 3 loaves = 6 ounces per loaf. For 7 loaves: 6 × 7 = 42 ounces. The equivalent ratio 18/3 = 42/7 confirms it.*